

NETWORK SECURITY ASSESSMENT REPORT

Metasploitable 2 — Authorized Vulnerability Assessment

Project: Network Security Assessment — Project 1

Assessment Type: Authorized Vulnerability Assessment / Security Testing

Target: Metasploitable 2

Target IP: 192.168.27.129

Assessment Platform: Kali Linux

Kali Linux IP: 192.168.27.128

Network: VMware VMnet1 Host-Only Network — 192.168.27.0/24

Assessment Year: 2026

Author: Cybersecurity Portfolio Project

1. Executive Summary

An authorized network security assessment was conducted against an intentionally vulnerable Metasploitable 2 virtual machine within an isolated VMware Host-Only network.

The objective of the assessment was to identify exposed network services, enumerate service versions, validate selected security weaknesses, analyze network traffic, assess associated risks, and provide appropriate remediation recommendations.

The assessment identified multiple security weaknesses across web services, file-sharing protocols, remote services, and cryptographic configuration.

The most significant confirmed finding was **unauthenticated WebDAV file creation and deletion**, where a file could be created, retrieved, and deleted without authentication.

Additional findings included:

- Broad NFS export of the root filesystem
- Anonymous SMB access to the tmp share
- Anonymous FTP authentication
- Cleartext FTP authentication traffic
- SMB signing not being required
- SMTP SSLv2 and weak cryptographic support
- Application information disclosure
- Legacy software and excessive network attack surface

Testing was restricted to the authorized Metasploitable 2 laboratory environment. No production systems or unauthorized external systems were targeted.

2. Assessment Objectives

The primary objectives of the assessment were to:

1. Identify active hosts within the authorized laboratory network.
2. Identify exposed TCP services and ports.
3. Enumerate service versions and technologies.
4. Identify potentially insecure service configurations.
5. Validate selected security weaknesses.
6. Capture and analyze network traffic using Wireshark.
7. Assess the potential impact of identified findings.
8. Develop practical remediation recommendations.
9. Produce professional security assessment documentation.

3. Scope and Authorization

3.1 Target

System: Metasploitable 2

IP Address: 192.168.27.129

3.2 Assessment Host

Operating System: Kali Linux

IP Address: 192.168.27.128

3.3 Network Environment

The assessment was conducted on an isolated VMware Host-Only network.

Network: 192.168.27.0/24

VMware Network: VMnet1 Host-Only

The isolated environment was used to ensure that testing activities remained within the controlled laboratory environment.

3.4 Authorization

The target was an intentionally vulnerable Metasploitable 2 virtual machine configured for security training and authorized testing.

No production systems or unauthorized external systems were targeted.

4. Assessment Methodology

The assessment followed a structured network security testing methodology.

Phase 1 — Lab Validation

The laboratory network configuration and connectivity between Kali Linux and Metasploitable 2 were verified.

Phase 2 — Host Discovery

Network connectivity and target availability were confirmed using ICMP and network discovery techniques.

Phase 3 — Port Scanning

Nmap was used to identify exposed TCP ports.

Phase 4 — Service Enumeration

Service and version detection was performed to identify applications, protocols, and software versions.

Phase 5 — Default Script Enumeration

Nmap default scripts were used to obtain additional information about exposed services and configurations.

Phase 6 — Protocol and Service Testing

Selected services were manually assessed, including:

- FTP
- SMB
- NFS
- HTTP
- WebDAV
- SMTP
- Tomcat

Phase 7 — Network Traffic Analysis

Wireshark was used to capture and analyze network traffic generated during testing.

Phase 8 — Risk Assessment

Confirmed findings were assessed based on their potential impact and likelihood of abuse within an appropriately configured environment.

Phase 9 — Reporting

Findings were documented with evidence, impact, severity, and recommended remediation

5. Tools Used

The following tools were used during the assessment:

- Kali Linux
- Nmap
- Wireshark
- smbclient
- showmount
- curl
- FTP client
- Linux networking utilities

6. Network Architecture

The laboratory consisted of two primary virtual machines.

Kali Linux

IP: 192.168.27.128

Used as the security assessment workstation.

Metasploitable 2

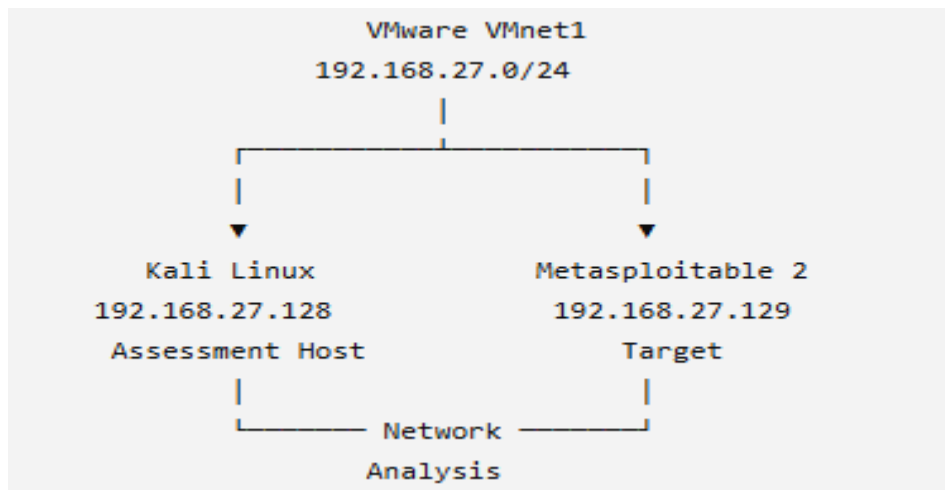
IP: 192.168.27.129

Used as the intentionally vulnerable assessment target.

Network

Both systems were connected through:

VMware VMnet1 Host-Only Network



7. Network Discovery

Connectivity between the assessment system and target was successfully established.

The target responded to network connectivity testing at:

192.168.27.129

This confirmed that the Metasploitable 2 system was reachable from the Kali Linux assessment environment.

8. Port and Service Enumeration

Nmap identified **23 open TCP ports** on the target.

Important services identified included:

Port Service / Version

21 FTP — vsftpd 2.3.4

22 SSH — OpenSSH 4.7p1

23 Telnet — Linux telnetd

25 SMTP — Postfix

53 DNS — BIND 9.4.2

80 HTTP — Apache 2.2.8

139 NetBIOS / Samba

445 SMB — Samba 3.0.20

111 RPC / rpcbind

2049 NFS

1524 Metasploitable root shell service

3306 MySQL 5.0.51a

5432 PostgreSQL 8.3

5900 VNC

6667 IRC — UnrealIRCd 3.2.8.1

8009 AJP13

8180 Apache Tomcat 5.5

The number of exposed services demonstrated a broad network attack surface.

9. Security Findings Summary

ID	Finding	Severity	Status
F-01	Unauthenticated WebDAV File Creation and Deletion	High	Confirmed
F-02	Broad NFS Export of Root Filesystem	High / Medium-High	Confirmed
F-03	Anonymous SMB Access to tmp Share	Medium	Confirmed
F-04	Anonymous FTP and Cleartext Authentication	Medium	Confirmed
F-05	SMB Signing Not Required	Medium	Confirmed
F-06	SMTP SSLv2 and Weak Cryptographic Support	Medium	Confirmed
F-07	Application Information Disclosure	Low-Medium	Confirmed
F-08	Legacy Software and Excessive Attack Surface	Low-Medium	Observation

10. Detailed Findings

F-01 — Unauthenticated WebDAV File Creation and Deletion

Severity: HIGH

Status: CONFIRMED

Description

The Apache WebDAV endpoint located at:

/dav/

permitted unauthenticated users to create and delete files.

The WebDAV service advertised multiple HTTP methods, including:

- PUT
- DELETE

- PROPFIND
- PROPPATCH
- COPY
- MOVE
- LOCK
- UNLOCK

Testing confirmed that a file could be created without authentication.

Evidence

A test file was created using:

PUT /dav/project1-test.txt

The server returned:

HTTP/1.1 201 Created

The file was subsequently retrieved successfully:

GET /dav/project1-test.txt

HTTP/1.1 200 OK

The file was then deleted:

DELETE /dav/project1-test.txt

HTTP/1.1 204 No Content

A subsequent request returned:

HTTP/1.1 404 Not Found

Impact

An unauthenticated attacker could potentially:

- Create unauthorized files
- Modify accessible web content
- Delete accessible files
- Abuse the WebDAV service as an attack surface
- Potentially facilitate further attacks depending on server configuration

Remote code execution was **not** claimed or demonstrated during this assessment.

Recommendation

- Disable WebDAV if it is not required.

- Require authentication and authorization.
- Restrict permitted HTTP methods.
- Apply least-privilege permissions.
- Separate writable directories from executable web content.
- Monitor WebDAV activity and file modifications.

F-02 — Broad NFS Export of Root Filesystem

Severity: HIGH / MEDIUM-HIGH

Status: CONFIRMED

Description

The target exposed an NFS export for the root filesystem:

/

The export was available to:

*

This represents an overly broad filesystem export.

Evidence

The following command was used:

```
showmount -e 192.168.27.129
```

The result was:

Export list for 192.168.27.129:

/ *

Nmap enumeration also identified NFS services on the target.

Impact

Broad NFS exports can expose sensitive filesystem resources to unauthorized systems.

Depending on export options and filesystem permissions, an attacker may potentially:

- Access sensitive files
- Read configuration information
- Access application data
- Access credentials stored in files
- Modify files where write permissions exist

Root access was **not** claimed solely from the NFS configuration.

Recommendation

- Do not export the root filesystem.
- Restrict NFS exports to trusted hosts or networks.
- Use read-only exports where possible.
- Configure appropriate root squashing.
- Apply least-privilege filesystem permissions.
- Disable NFS where it is not required.

F-03 — Anonymous SMB Access to tmp Share

Severity: MEDIUM

Status: CONFIRMED

Description

The SMB service permitted anonymous authentication and access to the tmp share without requiring credentials.

Evidence

The following command was used:

```
smbclient -L //192.168.27.129 -N
```

The target responded:

Anonymous login successful

The following shares were identified:

print\$

tmp

opt

IPC\$

ADMIN\$

Anonymous access to the tmp share was confirmed using:

```
smbclient //192.168.27.129/tmp -N
```

Directory contents were visible.

Impact

Anonymous SMB access can expose files and system resources to unauthenticated users.

Potential risks include:

- Unauthorized information disclosure
- Exposure of temporary files
- Increased attack surface
- Potential abuse if write permissions are available

Write access was not claimed or demonstrated.

Recommendation

- Disable guest/anonymous SMB access where unnecessary.
- Restrict share permissions.
- Remove unnecessary SMB shares.
- Require authentication for sensitive resources.
- Monitor anonymous SMB activity.

F-04 — Anonymous FTP and Cleartext Authentication

Severity: MEDIUM

Status: CONFIRMED

Description

The FTP service permitted anonymous authentication.

FTP authentication and commands can also be transmitted without encryption when TLS is not used.

Evidence

Anonymous authentication was successfully performed:

USER anonymous

331 Please specify the password.

PASS anonymous

230 Login successful.

Wireshark traffic analysis confirmed that FTP authentication commands were transmitted in cleartext.

Impact

An attacker capable of monitoring the network could potentially capture:

- FTP usernames
- FTP passwords
- Session information
- Other FTP commands

Anonymous access can also expose files or resources that should not be publicly accessible.

Recommendation

- Disable anonymous FTP unless explicitly required.
- Replace FTP with SFTP or another encrypted file-transfer protocol.
- If FTP must be retained, enforce TLS.
- Restrict FTP access through network controls.
- Monitor authentication activity.

F-05 — SMB Signing Not Required

Severity: MEDIUM

Status: CONFIRMED

Description

SMB enumeration identified that message signing was disabled or not required.

Evidence

Nmap SMB enumeration reported that SMB message signing was:
disabled / not required

Impact

When SMB signing is not required, certain man-in-the-middle or relay-style attacks may become more feasible when other environmental conditions are present.

No SMB relay attack was performed during this assessment.

Recommendation

- Require SMB message signing.
- Apply secure SMB configuration policies.
- Disable outdated SMB protocols where possible.
- Monitor unusual SMB authentication activity.

F-06 — SMTP SSLv2 and Weak Cryptographic Support

Severity: MEDIUM

Status: CONFIRMED

Description

The SMTP service supported legacy SSLv2 functionality and weak cryptographic options.

Evidence

Nmap enumeration identified:

SMTP

Postfix

SSLv2 supported

Weak SSLv2 ciphers

Impact

Legacy cryptographic protocols and weak cipher configurations can reduce the confidentiality and integrity of encrypted communications.

Recommendation

- Disable SSLv2.
- Disable obsolete cryptographic protocols.
- Require modern TLS.
- Use strong cipher suites.
- Regularly review mail-server TLS configuration.

F-07 — Application Information Disclosure

Severity: LOW-MEDIUM

Status: CONFIRMED

Description

The Mutillidae web application exposed internal application information through client-visible HTML source code.

The application disclosed information including:

Version: 2.1.19

PHP Version: 5.2.4-2ubuntu5.10

HTML comments also referenced database configuration information.

Impact

Information disclosure can provide attackers with useful intelligence about:

- Application versions
- Backend technologies
- Configuration
- Potential attack paths

The assessment did **not** assume that the database password mentioned in the HTML comment was valid.

Recommendation

- Remove sensitive comments from production HTML.
- Avoid exposing configuration information.
- Remove unnecessary version information.
- Keep applications and dependencies updated.
- Review source code before deployment.

F-08 — Legacy Software and Excessive Attack Surface

Severity: LOW-MEDIUM

Status: OBSERVATION

Description

The target exposed numerous legacy services and software versions.

Examples identified included:

vsftpd 2.3.4

OpenSSH 4.7p1

Apache 2.2.8

Samba 3.0.20

MySQL 5.0.51a

PostgreSQL 8.3

Apache Tomcat 5.5

UnrealIRCd 3.2.8.1

Impact

Legacy software may contain known vulnerabilities and may no longer receive security updates.

A large number of exposed services also increases the overall attack surface.

Recommendation

- Remove unnecessary services.
- Upgrade unsupported software.
- Apply security patches regularly.
- Restrict management interfaces.
- Use firewall rules to limit service exposure.
- Follow the principle of least functionality.

The presence of an old software version alone was not treated as proof of a specific exploitable vulnerability.

11. Security Observations

The following services were identified during the assessment but were not classified as confirmed vulnerabilities based solely on the observed behavior.

Tomcat Manager

The Tomcat Manager interface returned:

401 Unauthorized

Authentication was therefore required.

Tomcat Status

The Tomcat Status interface also returned:

401 Unauthorized

Tomcat Administration

The /admin/ interface presented an authentication page rather than providing unauthenticated administrative access.

phpMyAdmin

The phpMyAdmin interface was accessible but required authentication.

DVWA

The DVWA login interface was accessible but authentication was required.

These observations demonstrate that **service exposure does not automatically mean unauthorized access was achieved.**

12. Wireshark Traffic Analysis

Wireshark was used to capture and analyze network traffic generated during the assessment.

One of the most important observations was FTP authentication being transmitted in cleartext.

The captured traffic included:

USER anonymous

331 Please specify the password.

PASS anonymous

230 Login successful.

This provided packet-level evidence supporting the anonymous FTP and cleartext authentication finding.

Wireshark analysis demonstrated the importance of validating security findings at the network-protocol level rather than relying solely on service enumeration.

13. Risk Assessment

The overall assessment identified weaknesses across several security domains.

Authentication and Access Control

- Anonymous FTP authentication
- Anonymous SMB authentication
- Unauthenticated WebDAV file operations
- Broad NFS export

Network Security

- Excessive exposed services
- Legacy network services
- SMB signing not required

Cryptography

- SSLv2 support
- Weak SMTP cryptographic configuration
- Cleartext FTP authentication
-

Application Security

- WebDAV misconfiguration
- Application information disclosure
- Legacy web technologies

The highest priority should be given to access-control weaknesses that allow unauthenticated users to interact with filesystem or web resources.

14. Remediation Priorities

Priority 1 — Immediate

1. Disable or secure unauthenticated WebDAV access.
2. Remove the broad NFS root filesystem export.
3. Restrict anonymous SMB access.
4. Disable anonymous FTP.

Priority 2 — High

1. Require SMB signing.
2. Disable SSLv2 and weak cryptographic protocols.
3. Restrict unnecessary network services.
4. Apply appropriate firewall restrictions.

Priority 3 — Medium

1. Remove application information disclosure.
2. Upgrade unsupported software.
3. Review exposed services.
4. Establish regular vulnerability assessment and patch-management processes.

15. Recommended Security Improvements

Based on the findings, the following security improvements are recommended:

Access Control

- Enforce authentication for sensitive services.
- Disable unnecessary guest and anonymous access.
- Apply least-privilege permissions.
- Restrict network access to trusted hosts.

Network Hardening

- Disable unnecessary services.
- Implement host-based and network firewalls.
- Segment sensitive systems.
- Restrict management interfaces.

Secure Protocols

- Replace FTP with SFTP.
- Disable obsolete cryptographic protocols.
- Require secure TLS configurations.
- Require SMB signing.

Patch Management

- Maintain supported software versions.
- Apply security updates regularly.
- Track vulnerable or unsupported services.

Monitoring

- Monitor authentication activity.
- Monitor file modifications.
- Monitor unusual network traffic.
- Centralize security logs where possible.

16. Evidence and Project Artifacts

The assessment produced supporting technical artifacts, including:

- Nmap service enumeration results
- Nmap XML output
- Nmap GNMAP output
- Nmap default-script results
- Wireshark packet capture
- Wireshark traffic-analysis screenshots
- SMB enumeration results
- NFS enumeration results

- Web service testing evidence
- Security findings documentation
- This professional assessment report

The supporting files are organized within the project GitHub repository.

17. Skills Demonstrated

This project demonstrates practical experience with:

- Network reconnaissance
- Host discovery
- Nmap port scanning
- Service/version enumeration
- SMB enumeration
- NFS enumeration
- Web security assessment
- WebDAV security testing
- FTP security assessment
- Wireshark packet analysis
- Evidence collection
- Vulnerability identification
- Risk assessment
- Security remediation
- Technical documentation
- Professional security reporting

18. Conclusion

The assessment successfully demonstrated a structured network security assessment methodology against an intentionally vulnerable laboratory environment.

Multiple security weaknesses were identified and validated through a combination of automated enumeration and manual testing.

The assessment demonstrated the ability to progress from:

Reconnaissance → Enumeration → Validation → Evidence Collection → Risk Assessment → Remediation → Professional Reporting

The most significant confirmed finding was unauthenticated WebDAV file creation and deletion.

Additional findings demonstrated weaknesses involving NFS, SMB, FTP, SMTP cryptography, application information disclosure, and excessive service exposure.

The project demonstrates practical cybersecurity skills relevant to entry-level and junior roles such as:

- Junior Cybersecurity Analyst
- SOC Analyst
- Security Operations Analyst
- IT Security Analyst
- Junior Penetration Tester
- Vulnerability Assessment Analyst
- IT/System Security Administrator

All testing was performed against the authorized Metasploitable 2 virtual machine within an isolated laboratory network.

No production systems or unauthorized external systems were targeted.

19. Disclaimer

This report documents security testing performed in an intentionally vulnerable and isolated laboratory environment.

The techniques and findings described in this report should only be applied to systems for which explicit authorization has been granted.

Project Information

Project: Network Security Assessment — Project 1

Target: Metasploitable 2

Assessment Platform: Kali Linux

Environment: VMware Host-Only Network

Year: 2026

Project Type: Cybersecurity Portfolio Project

END OF REPORT

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